**Global Earthquake Dashboard**

**Capillary Technologies Internship — Assignment 2 | Prepared by: G.Jeswanth**

## Objective

This dashboard explores earthquake activity using real data from the United States Geological Survey (USGS). It presents earthquake events in an accessible, interactive interface so users can review locations, magnitudes, depths, and activity over time.

## Tools and Data Source

The application is built with Python and Streamlit. Requests retrieves data from the public USGS FDSN Event Web Service; Pandas organizes the records; Plotly creates charts; and PyDeck displays earthquake locations on an interactive map. The API returns GeoJSON and does not require an API key.

## Dashboard Features

The dashboard provides filters for the time period and minimum magnitude, summary cards for total events, strongest magnitude, average magnitude, and average depth, a world map with hover details, a magnitude-distribution chart, a daily activity chart, an earthquake records table with USGS event links, and CSV export. API request and data-processing errors are displayed to the user.

## LLM-Assisted Development

I used ChatGPT to break the task into manageable steps, plan the dashboard structure, generate and explain the Python/Streamlit code, and troubleshoot the API request. I checked the API response in a browser before connecting it to the application, then ran the app locally and tested its filters, map, charts, table, and CSV download. The code was refined based on observed results rather than relying on generated output alone.

## Verification

The USGS test request returned HTTP status 200 and a FeatureCollection containing 370 events for the selected query (10–17 October 2026; minimum magnitude 2.5). The response included magnitude, place, event time, coordinates, depth, and the USGS event URL. The dashboard subsequently loaded data and its main interactive features worked during local testing. Event totals may differ when filters or query dates change.

## Limitations

The dashboard requires an internet connection and depends on the availability and contents of the USGS service. Results are limited by the selected date range and minimum magnitude, and the displayed events should not be interpreted as every earthquake worldwide. Data is cached for five minutes, so the display may not reflect changes immediately. Data source:- https://earthquake.usgs.gov/fdsnws/event/1/